Geocomputation

Welcome



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Geocomputation

Module Overview



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Module aims

- Geocomputation provides you with an introduction to the principles of GIScience, spatial analysis, and the use of programming for spatial data analysis.
- Over the next ten weeks, you will learn about the theory, methods and tools of spatial analysis using the R programming language within the RStudio software environment.
- You will further learn how to find, clean, and manage demographic, socio-economic, and environmental datasets, and analyse them using core spatial analysis techniques.

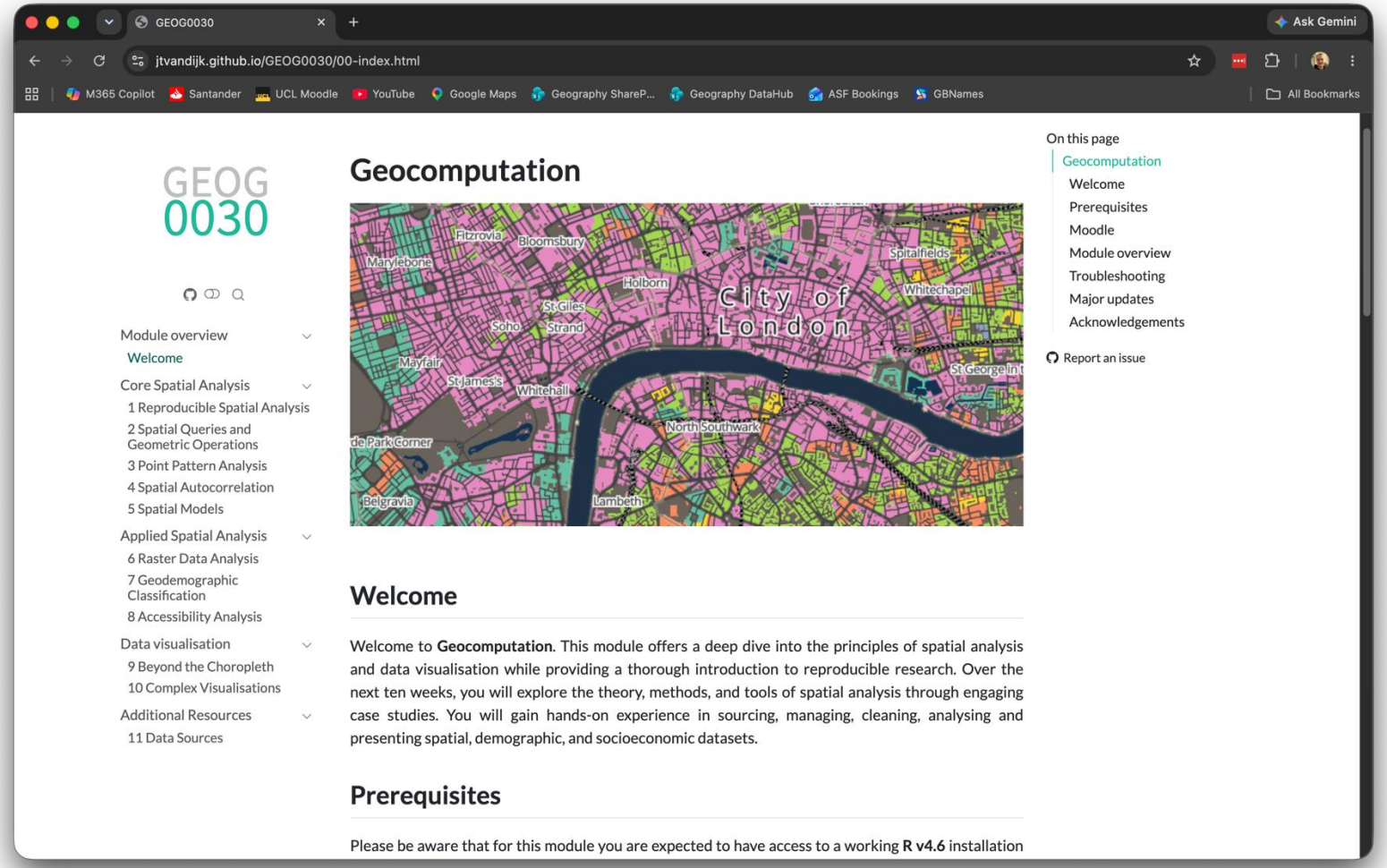
Module outline

W1	Reproducible Spatial Analysis	Core Spatial Analysis
W2	Spatial Queries and Geometric Operations	
W3	Point Pattern Analysis	
W4	Spatial Autocorrelation	
W5	Spatial Models	
W6	Raster Data Analysis	Applied Spatial Analysis
W7	Geodemographic Classification	
W8	Accessibility Analysis	
W9	Beyond the Choropleth	Data Visualisation
W10	Complex Visualisations	

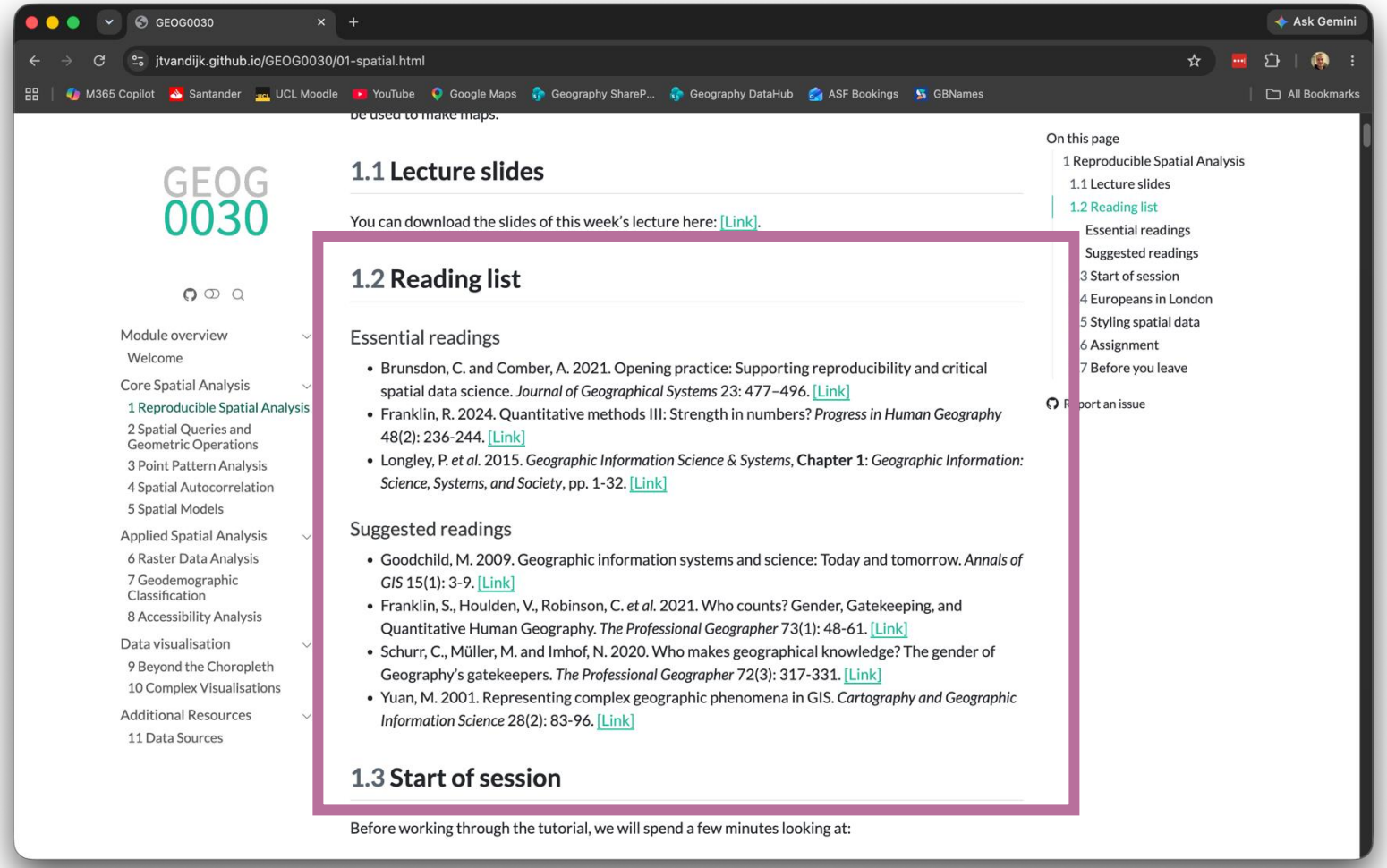
Module structure

- This module consists of ten lectures (Monday morning) and ten supervised computer tutorials (Group A: Tuesday morning, Group B: Tuesday afternoon).
- Each week will have its own reading list. Reading lists and instructions for the computer tutorial are published on the dedicated GEOG0030 Geocomputation Workbook. The link to this workbook is available on Moodle.

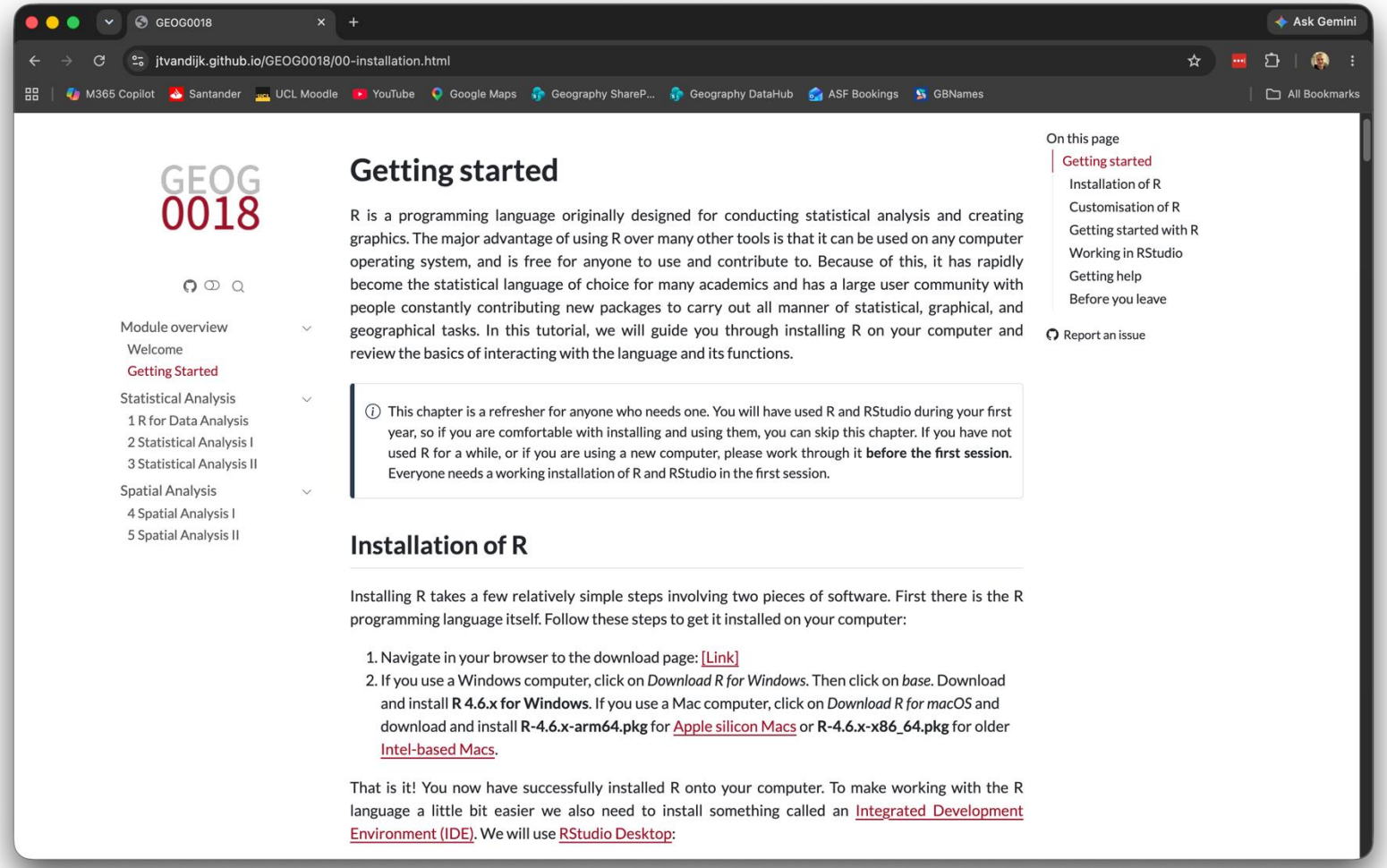
Workbook


GEOG0030
jtvandijk.github.io/GEOG0030/00-index.html
Ask Gemini
M365 Copilot
Santander
UCL Moodle
YouTube
Google Maps
Geography ShareP...
Geography DataHub
ASF Bookings
GBNames
All Bookmarks
GEOG 0030
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10 Complex Visualisations
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11 Data Sources
Geocomputation
Fitzrovia
Bloomsbury
Marylebone
Holborn
Spitalfields
Whitechapel
City of London
St Giles
Soho
Strand
Mayfair
St James's
Whitehall
North Southwark
Belgravia
Lambeth
Welcome
Welcome to Geocomputation. This module offers a deep dive into the principles of spatial analysis and data visualisation while providing a thorough introduction to reproducible research. Over the next ten weeks, you will explore the theory, methods, and tools of spatial analysis through engaging case studies. You will gain hands-on experience in sourcing, managing, cleaning, analysing and presenting spatial, demographic, and socioeconomic datasets.
Prerequisites
Please be aware that for this module you are expected to have access to a working R v4.6 installation
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Workbook


GEOG0030
jtvandijk.github.io/GEOG0030/01-spatial.html
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be used to make maps.
GEOG 0030
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Additional Resources
11 Data Sources
1.1 Lecture slides
You can download the slides of this week's lecture here: [Link].
1.2 Reading list
Essential readings
Brunsdon, C. and Comber, A. 2021. Opening practice: Supporting reproducibility and critical spatial data science. Journal of Geographical Systems 23: 477–496. [Link]
Franklin, R. 2024. Quantitative methods III: Strength in numbers? Progress in Human Geography 48(2): 236-244. [Link]
Longley, P. et al. 2015. Geographic Information Science & Systems, Chapter 1: Geographic Information: Science, Systems, and Society, pp. 1-32. [Link]
Suggested readings
Goodchild, M. 2009. Geographic information systems and science: Today and tomorrow. Annals of GIS 15(1): 3-9. [Link]
Franklin, S., Houlden, V., Robinson, C. et al. 2021. Who counts? Gender, Gatekeeping, and Quantitative Human Geography. The Professional Geographer 73(1): 48-61. [Link]
Schurr, C., Müller, M. and Imhof, N. 2020. Who makes geographical knowledge? The gender of Geography's gatekeepers. The Professional Geographer 72(3): 317-331. [Link]
Yuan, M. 2001. Representing complex geographic phenomena in GIS. Cartography and Geographic Information Science 28(2): 83-96. [Link]
1.3 Start of session
Before working through the tutorial, we will spend a few minutes looking at:
On this page
1 Reproducible Spatial Analysis
1.1 Lecture slides
1.2 Reading list
Essential readings
Suggested readings
3 Start of session
4 Europeans in London
5 Styling spatial data
6 Assignment
7 Before you leave
Report an issue

Refresher


GEOG0018
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Module overview
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2 Statistical Analysis I
3 Statistical Analysis II
Spatial Analysis
4 Spatial Analysis I
5 Spatial Analysis II
Getting started
R is a programming language originally designed for conducting statistical analysis and creating graphics. The major advantage of using R over many other tools is that it can be used on any computer operating system, and is free for anyone to use and contribute to. Because of this, it has rapidly become the statistical language of choice for many academics and has a large user community with people constantly contributing new packages to carry out all manner of statistical, graphical, and geographical tasks. In this tutorial, we will guide you through installing R on your computer and review the basics of interacting with the language and its functions.
This chapter is a refresher for anyone who needs one. You will have used R and RStudio during your first year, so if you are comfortable with installing and using them, you can skip this chapter. If you have not used R for a while, or if you are using a new computer, please work through it before the first session. Everyone needs a working installation of R and RStudio in the first session.
Installation of R
Installing R takes a few relatively simple steps involving two pieces of software. First there is the R programming language itself. Follow these steps to get it installed on your computer:
1. Navigate in your browser to the download page: [Link]
2. If you use a Windows computer, click on Download R for Windows. Then click on base. Download and install R 4.6.x for Windows. If you use a Mac computer, click on Download R for macOS and download and install R-4.6.x-arm64.pkg for Apple silicon Macs or R-4.6.x-x86_64.pkg for older Intel-based Macs.
That is it! You now have successfully installed R onto your computer. To make working with the R language a little bit easier we also need to install something called an Integrated Development Environment (IDE). We will use RStudio Desktop:
On this page
Getting started
Installation of R
Customisation of R
Getting started with R
Working in RStudio
Getting help
Before you leave
Report an issue

Practicalities

- Lecture slides will be made available on the GEOG0030 Geocomputation workbook webpage before each lecture. Lectures are recorded using Panopto.
- During the computer tutorials you are expected to work independently through the assignments in the workbook with the opportunity to ask questions.

Communication

- All important information will be communicated through Moodle.
- For specific questions on the module, my Academic Support and Feedback (ASF) Office Hours are scheduled on TBC and TBC. Slots can be booked through Microsoft Bookings: [Link]

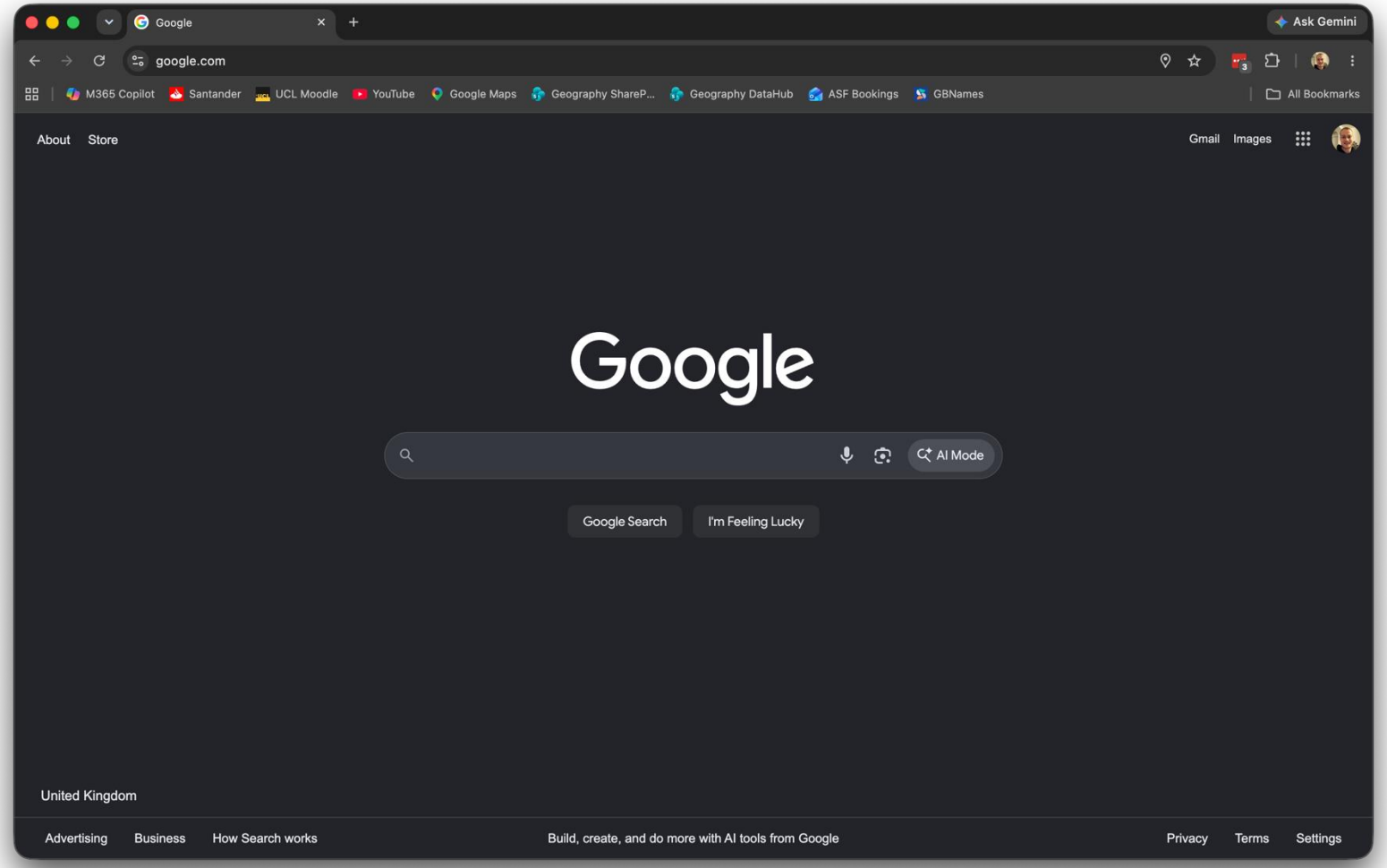
Assessment

- Social Atlas Coursework Assessment (60%): The first assessment will involve the completion of a spatial analysis project, based on the theory, concepts and application learnt during the module. For this coursework you are required to create a small *social atlas* on a topic or area that interests you. Deadline: TBC.
- Specific Coursework guidance on Moodle (including past examples with indicative grades), further attention to this in W05's Lecture.
- Exam Assessment (40%): The second assessment will take the form of a written two-hour Exam, further attention to this in W10's Lecture.

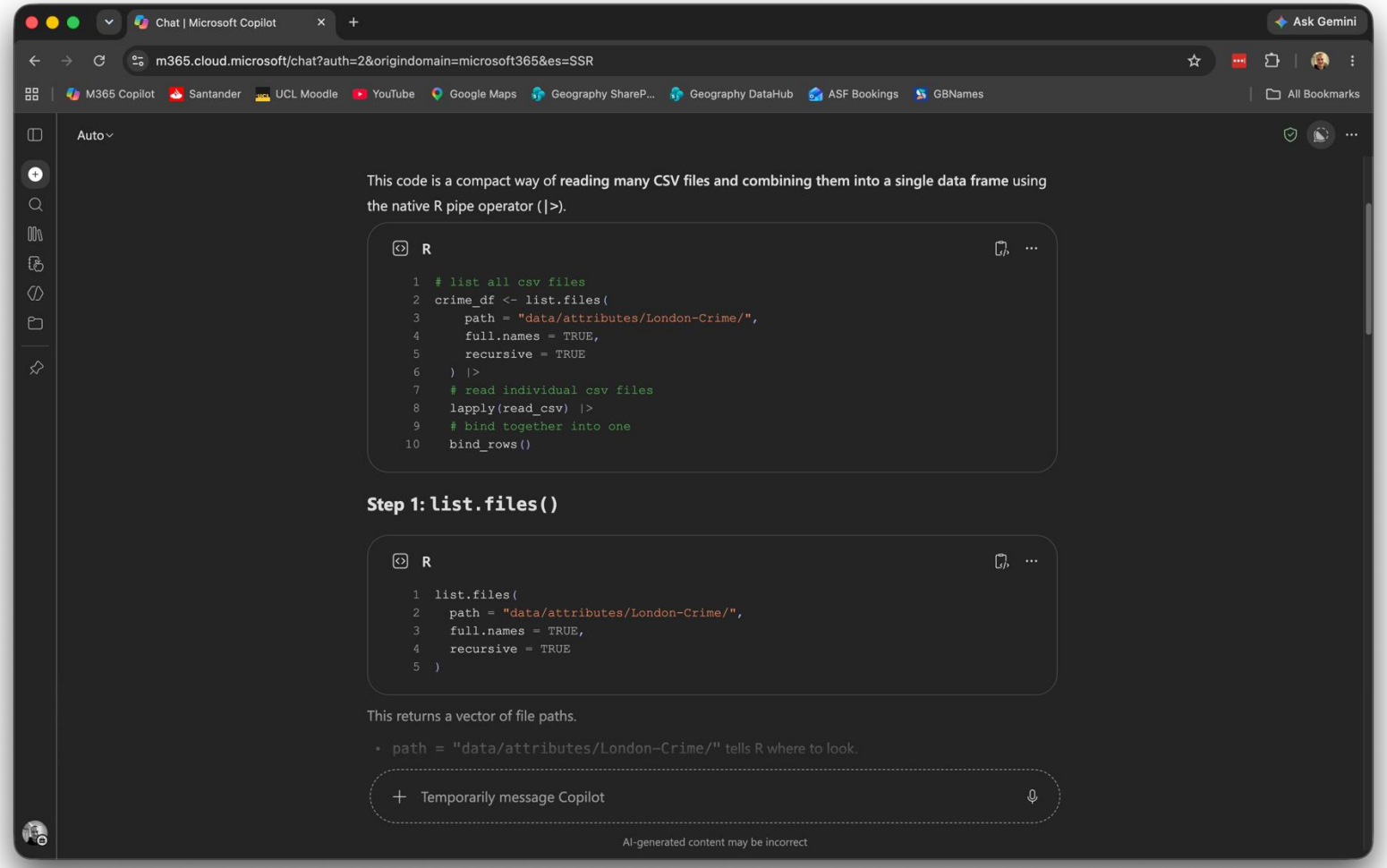
Troubleshooting

- Refresher GEOG0018 Methods in Human Geography.
- Ask a question at the end of a plenary lecture.
- Ask a question during a computer tutorial.
- Drop by the Departmental Coding Therapy sessions.

Troubleshooting



Troubleshooting



Coding and AI in this module

- Weeks 1 to 8: you write the code yourself. Making and fixing your own mistakes is how you learn to spot them.
- Weeks 9 and 10: you work with Microsoft Copilot, the AI assistant that UCL provides. Your task is to check what it produces, not to accept it.
- AI can return code that runs and still gives the wrong answer, for example a wrong projection or duplicated rows.

Questions

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Geocomputation

Reproducible Spatial Analysis



Module outline

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This week

- What is GIScience?
- What is Geocomputation?

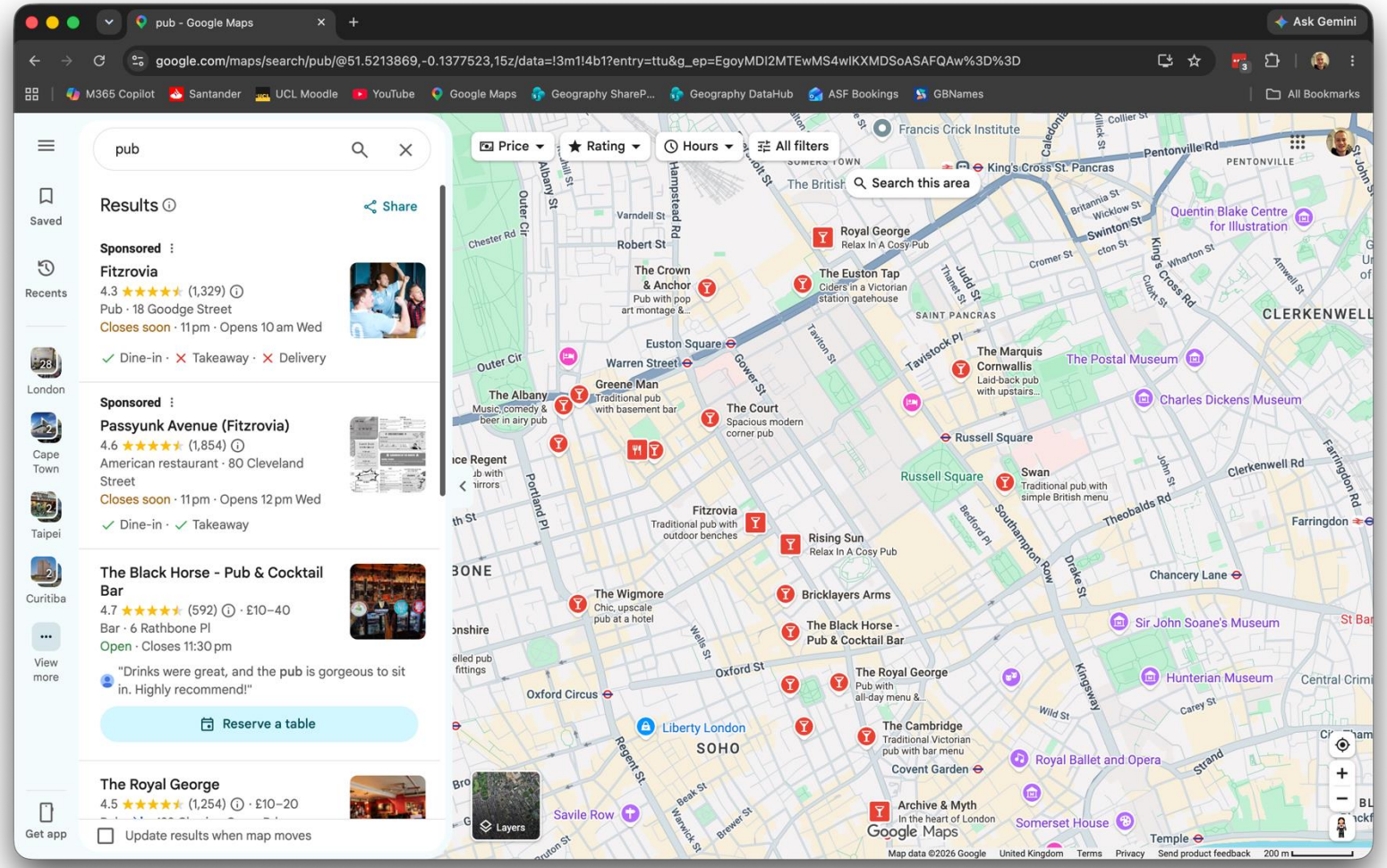
Before we start

- Go to www.menti.com
- Use code: XXXX XXXX

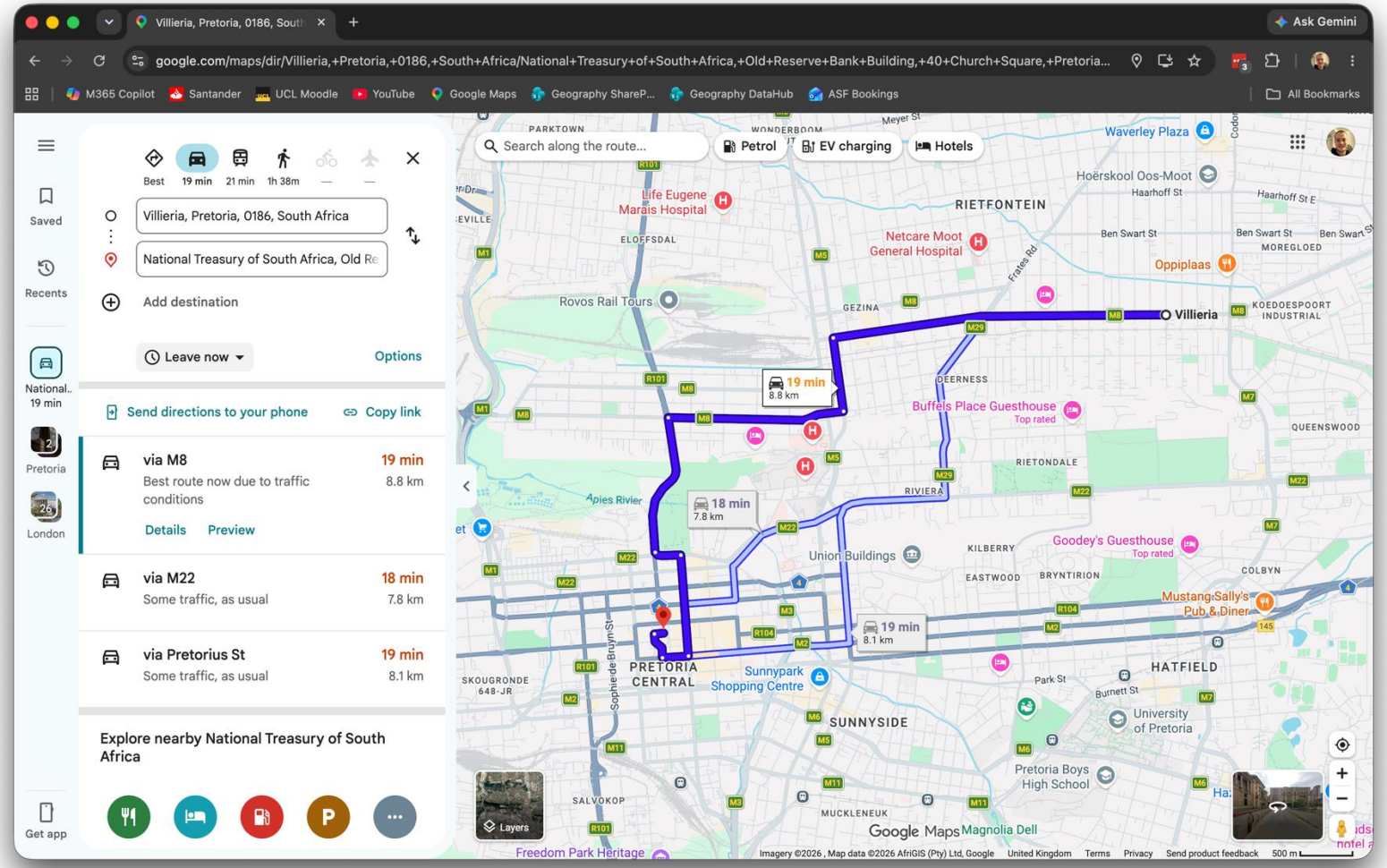
GIScience

- The Geocomputation module is designed to provide you with an introduction to the principles of GIS, spatial analysis and the use of programming for data analysis.
- GIS: Geographic Information Systems / Geographic Information Science
- Why do we need a science of Geographic Information?

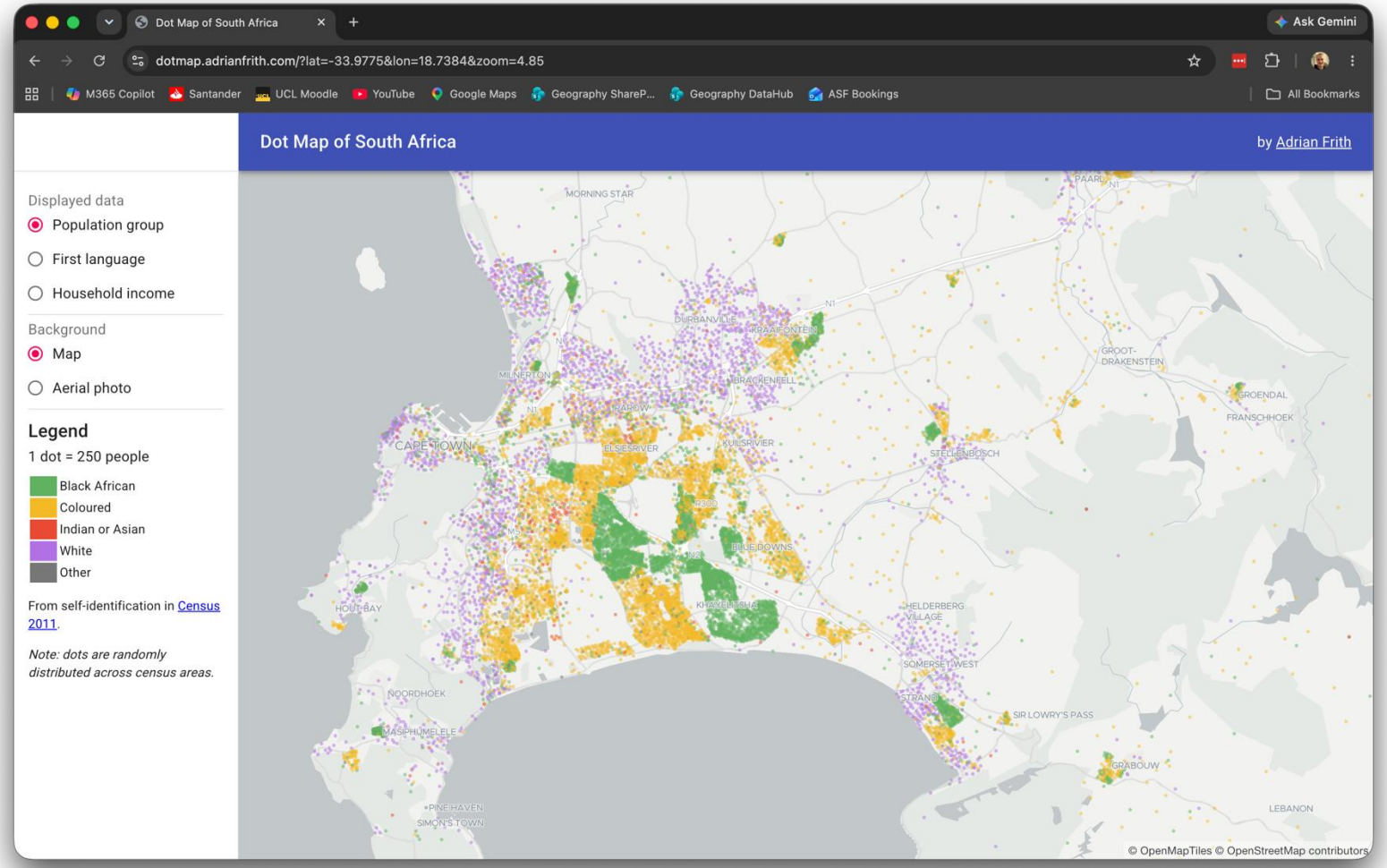
Pubs in Bloomsbury



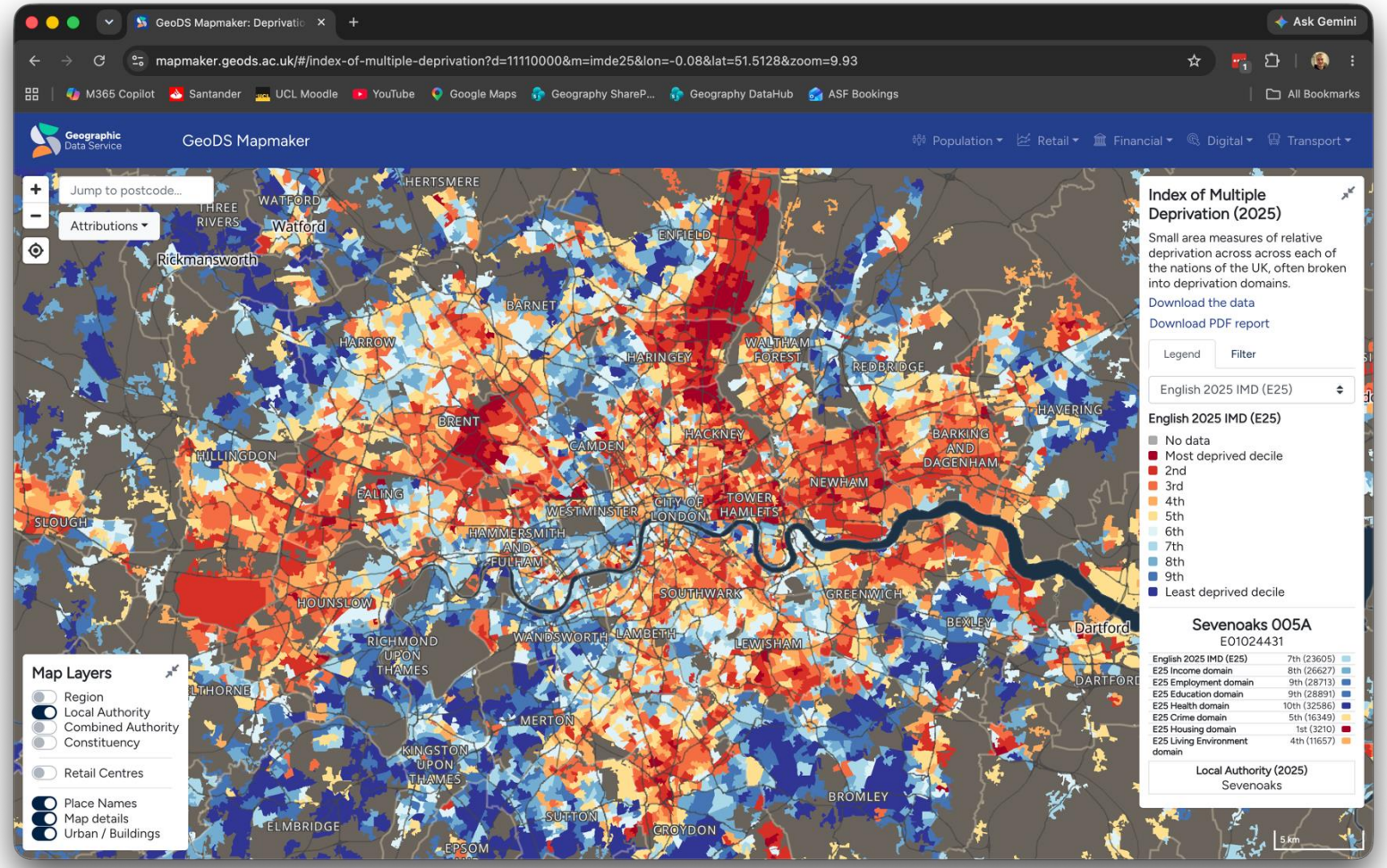
Finding your way abroad



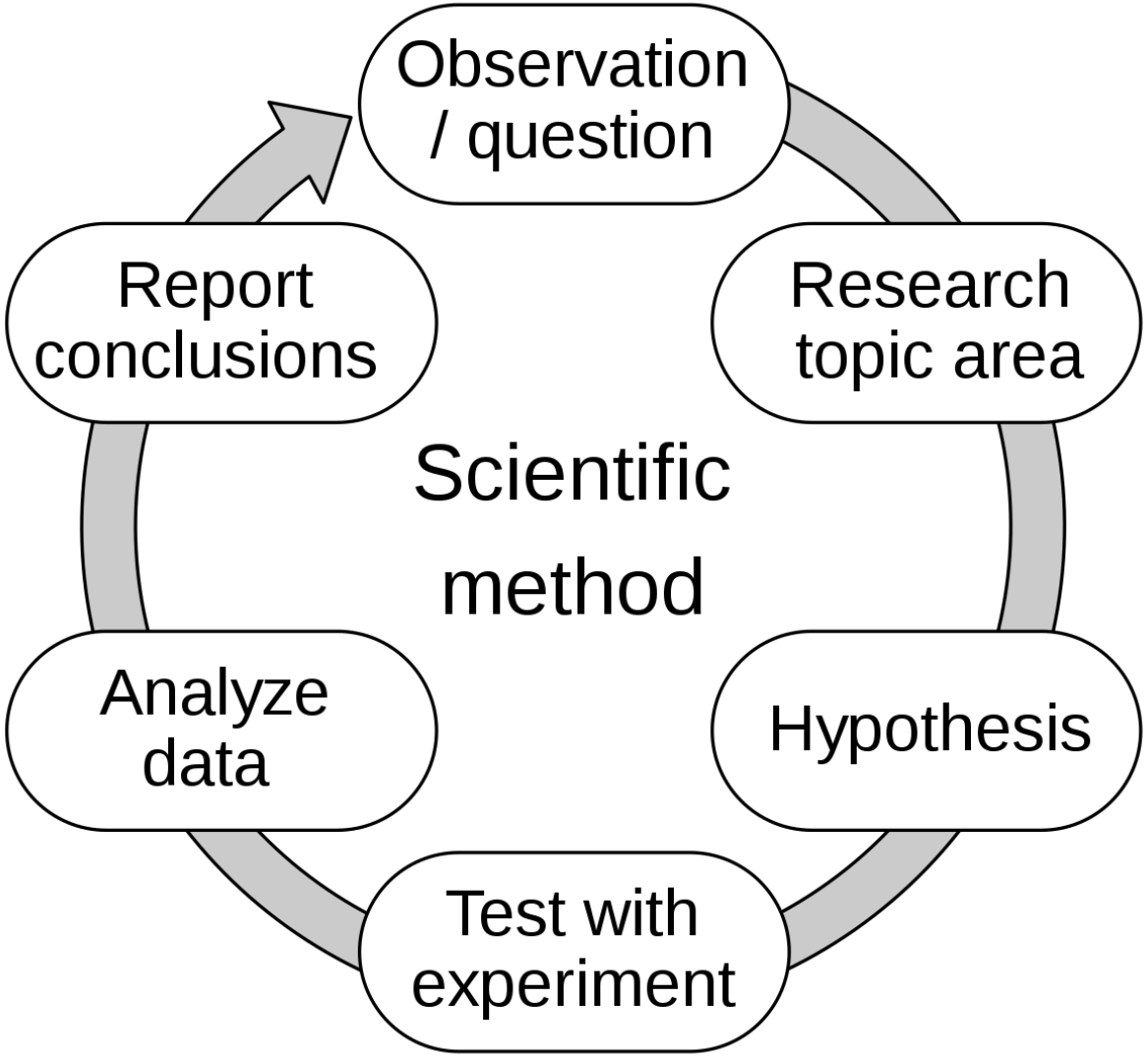
Segregation in Cape Town



Index of Multiple Deprivation in England



Foundations of GIScience



Why spatial is special

- Nearby places tend to be more alike than distant places: spatial dependence.
- Relationships can differ from place to place: spatial heterogeneity.

Foundations of GIScience

- Spatial information requires specialised methods and tools to handle its unique properties. To work effectively with spatial data, we need to:
  - Represent spatial data accurately in a digital format.
  - Store and organise spatial data for easy access and interaction.
  - Perform analyses to extract insights from spatial data.
  - Communicate results effectively.
- Together they constitute the four ‘pillars’ of GIScience, the science behind GIS software.

Foundations of GIScience

(1) Spatial modelling and digital representation:

How to represent spatial phenomena digitally?

(2) Geographic Information Systems:

How to store, manage, retrieve and query these digital spatial phenomena.

Foundations of GIScience

(3) Spatial analysis:

The theory, principles, and techniques that enable analysis of spatial data to look at spatial patterns, processes, and relationships.

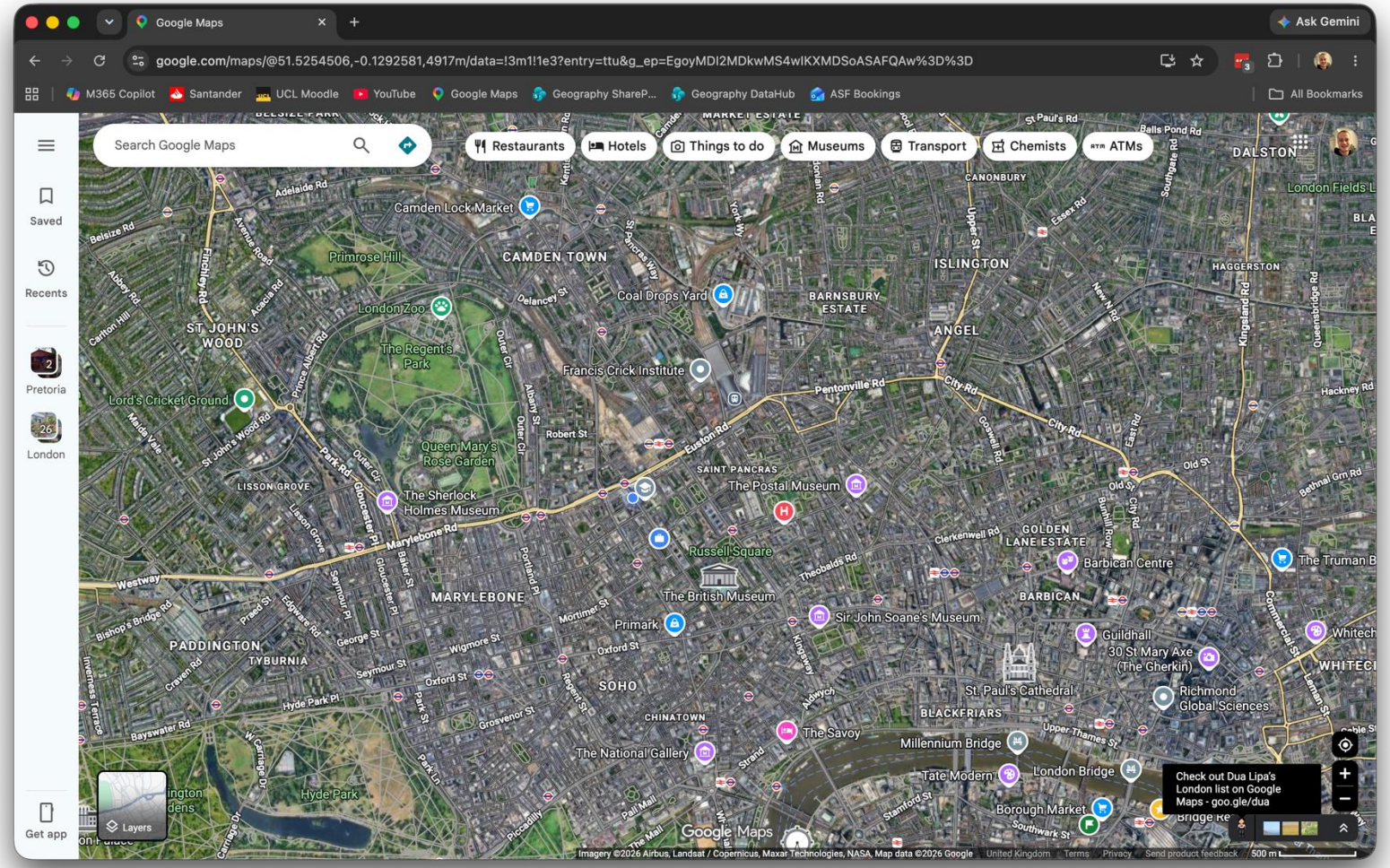
(4) Cartography and visualisation:

How to present spatial data and the results from spatial analysis to communicate results (including projections and map conventions).

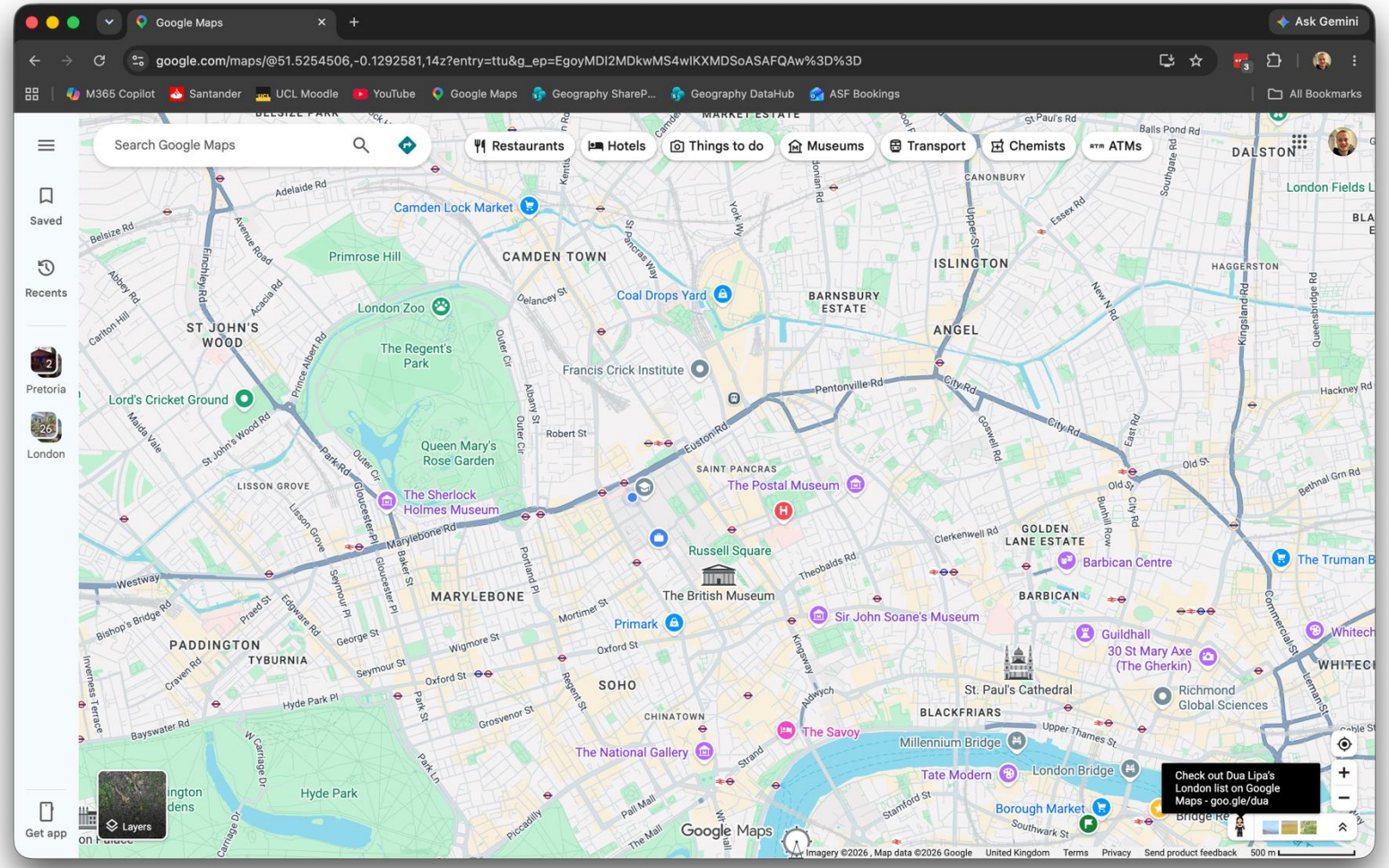
Spatial modelling and digital representation



Spatial modelling and digital representation



Spatial modelling and digital representation



Spatial modelling and digital representation

- GIScience relies on representing spatial information in a digital format.
- Traditionally, geographic information is conveyed in two primary ways:

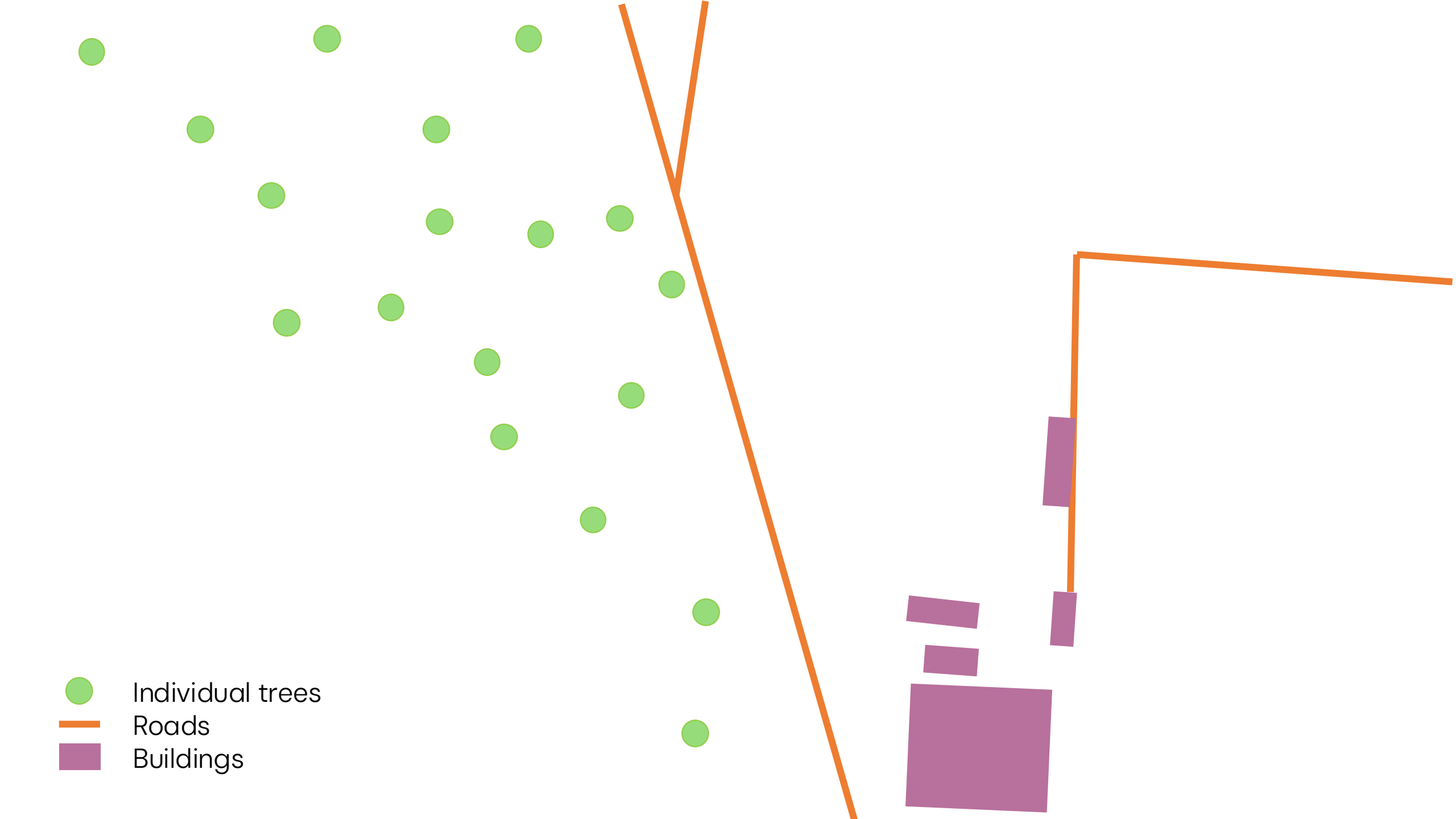
  Vector This method uses a finite set of discrete geometric objects, such as points, lines, and polygons, to represent spatial features.

  Raster This approach employs images or grids to represent surfaces, with each cell or pixel holding a value, often indicating attributes like colours or measurements.
- Each feature also has attributes: a row in a table with information about it, such as a road name or a population count.

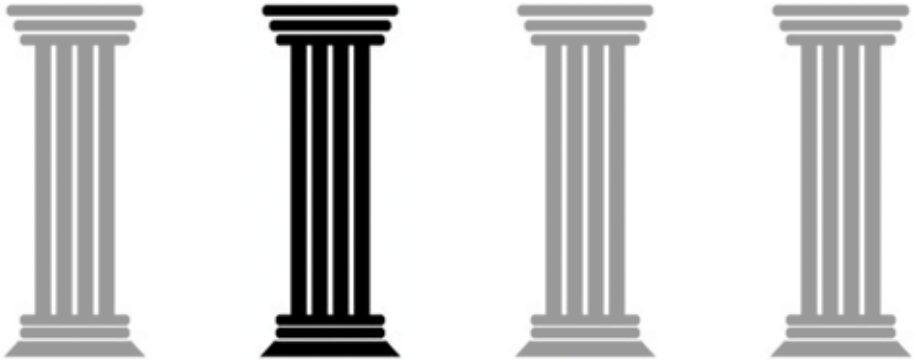







Individual trees
Roads
Buildings

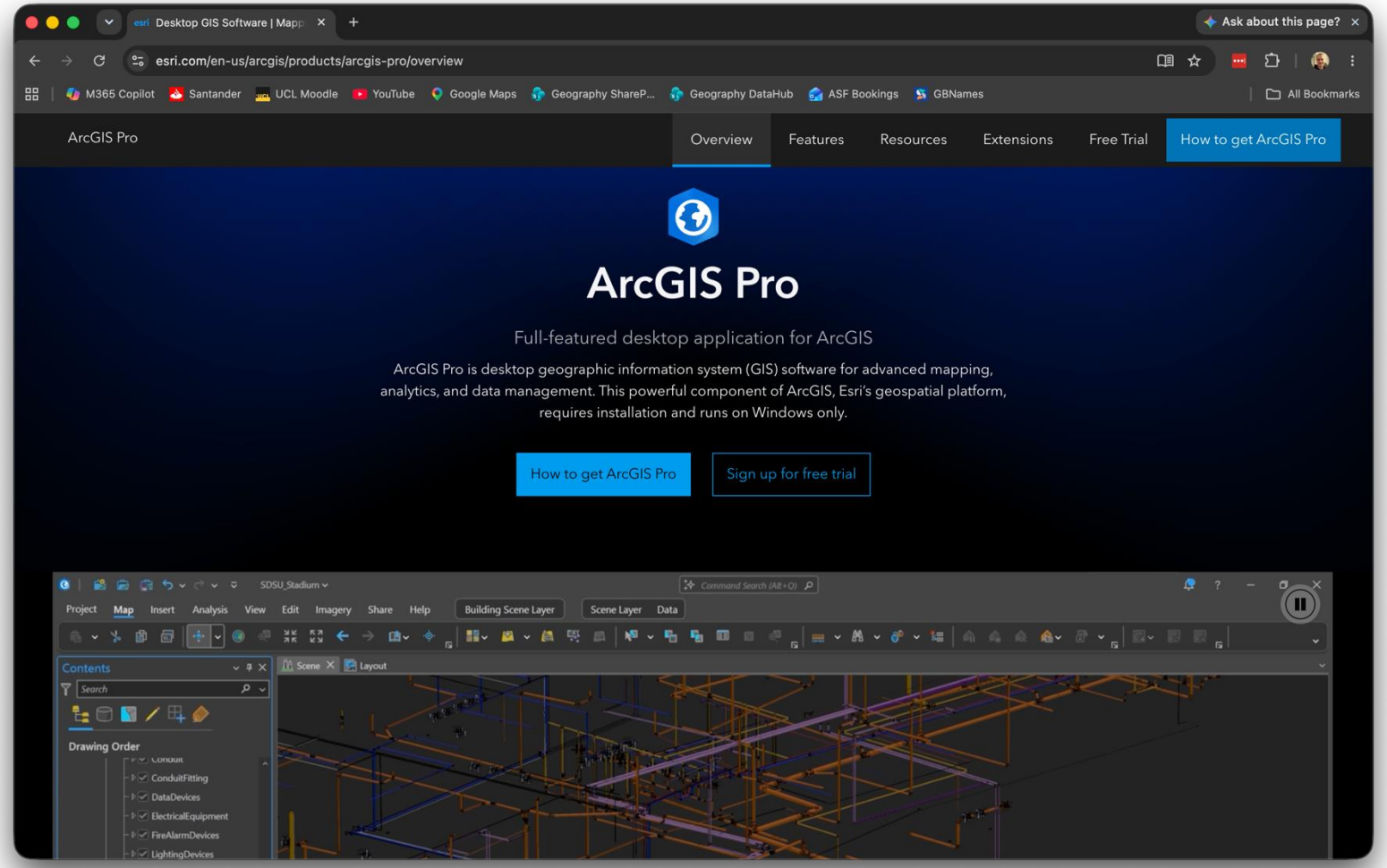
Geographic Information Systems



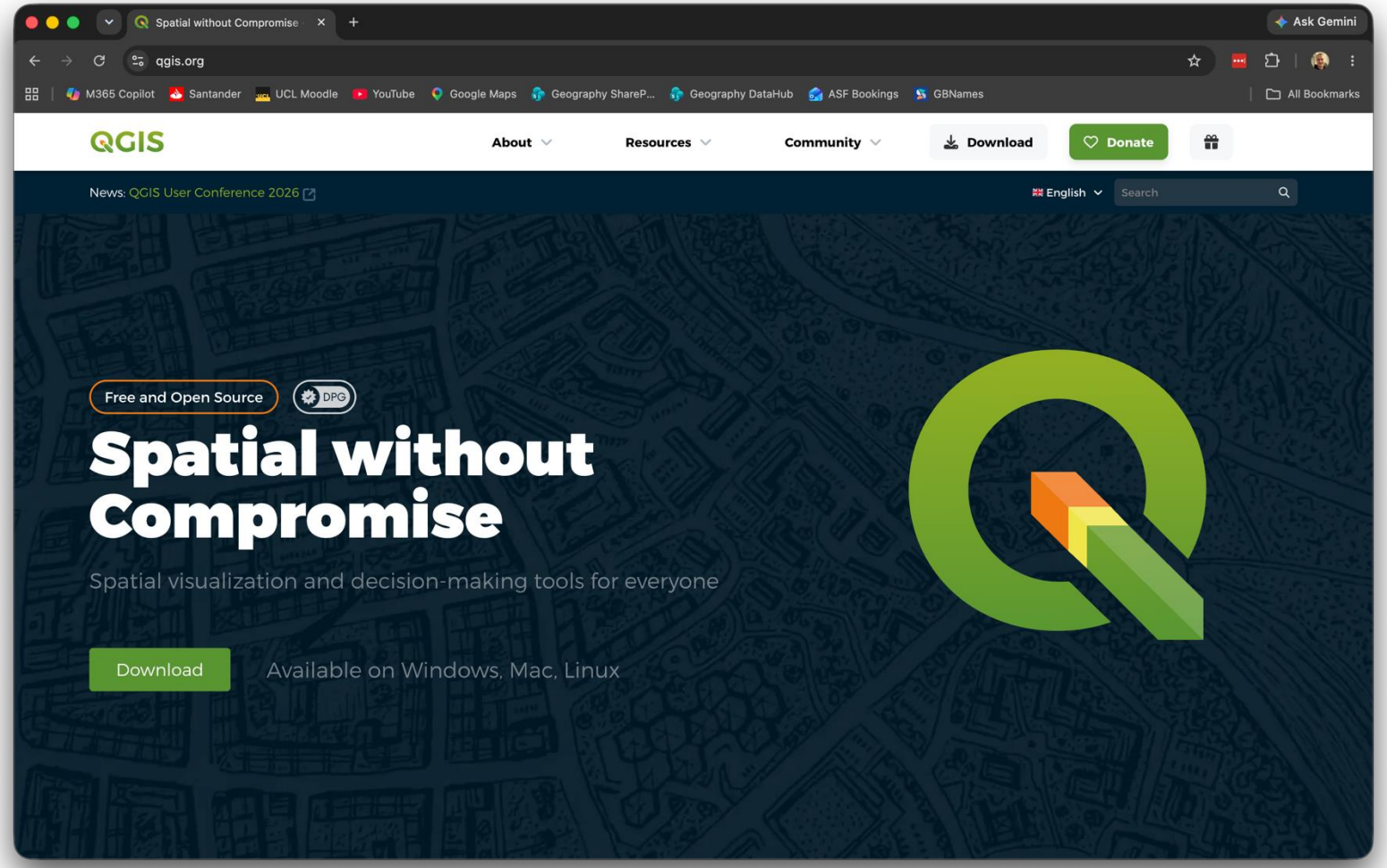
Geographic Information Systems

- GI systems help us to manage spatial data: organisation, storage, access and retrieval, and manipulation.
- Dedicated GIS software: ArcGIS Pro, QGIS, GeoDa
- Software that has GIS functionality: R, Python

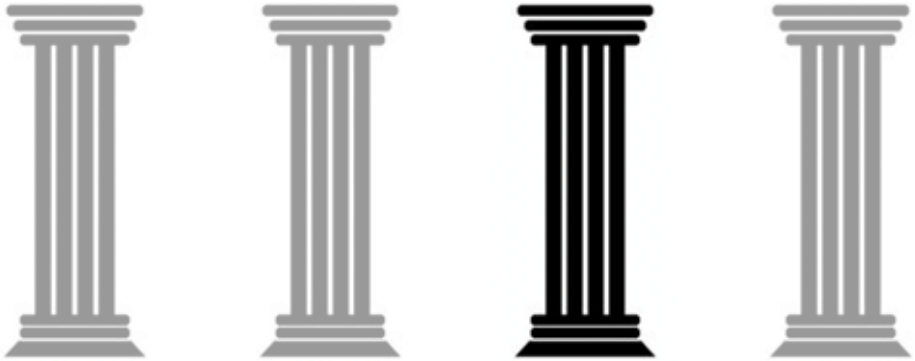
Geographic Information Systems



Geographic Information Systems



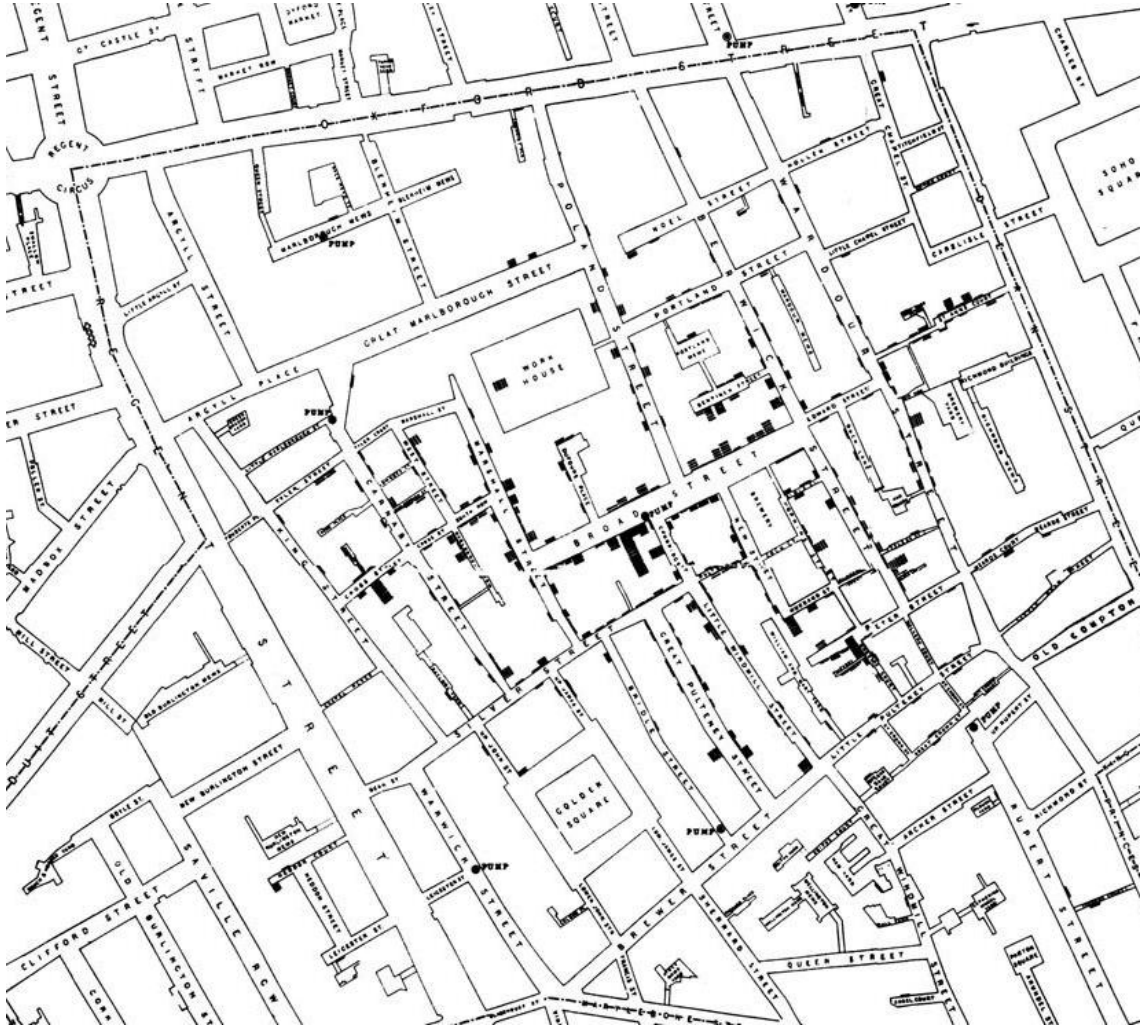
Spatial analysis



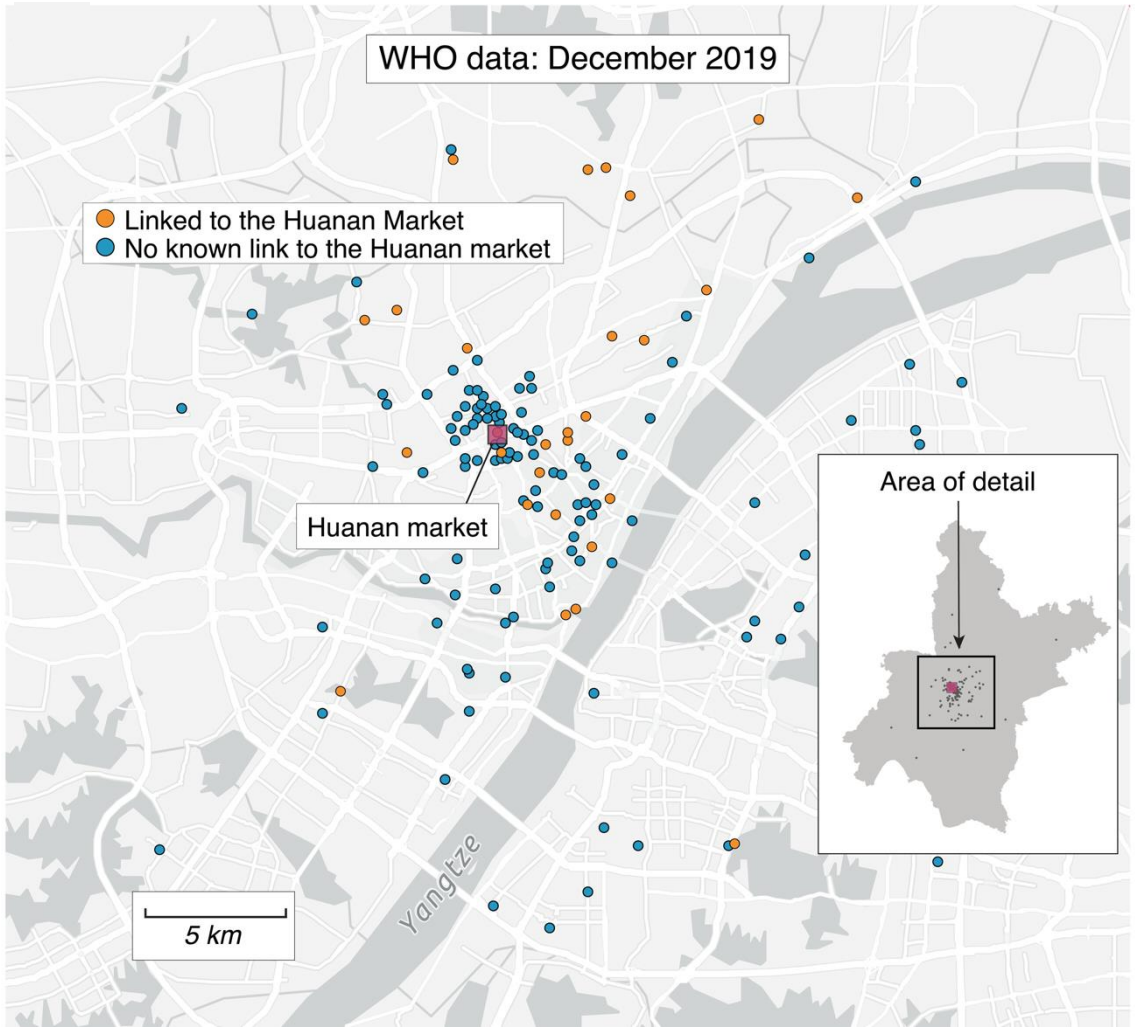
Spatial analysis

- The theory, principles, and techniques that enable analysis of spatial data to discover spatial patterns, processes and relationships.
- Spatial analysis seeks to enhance our understanding of the world by transforming data into meaningful information, quantifying elements such as distributions and spatial processes.

Spatial analysis

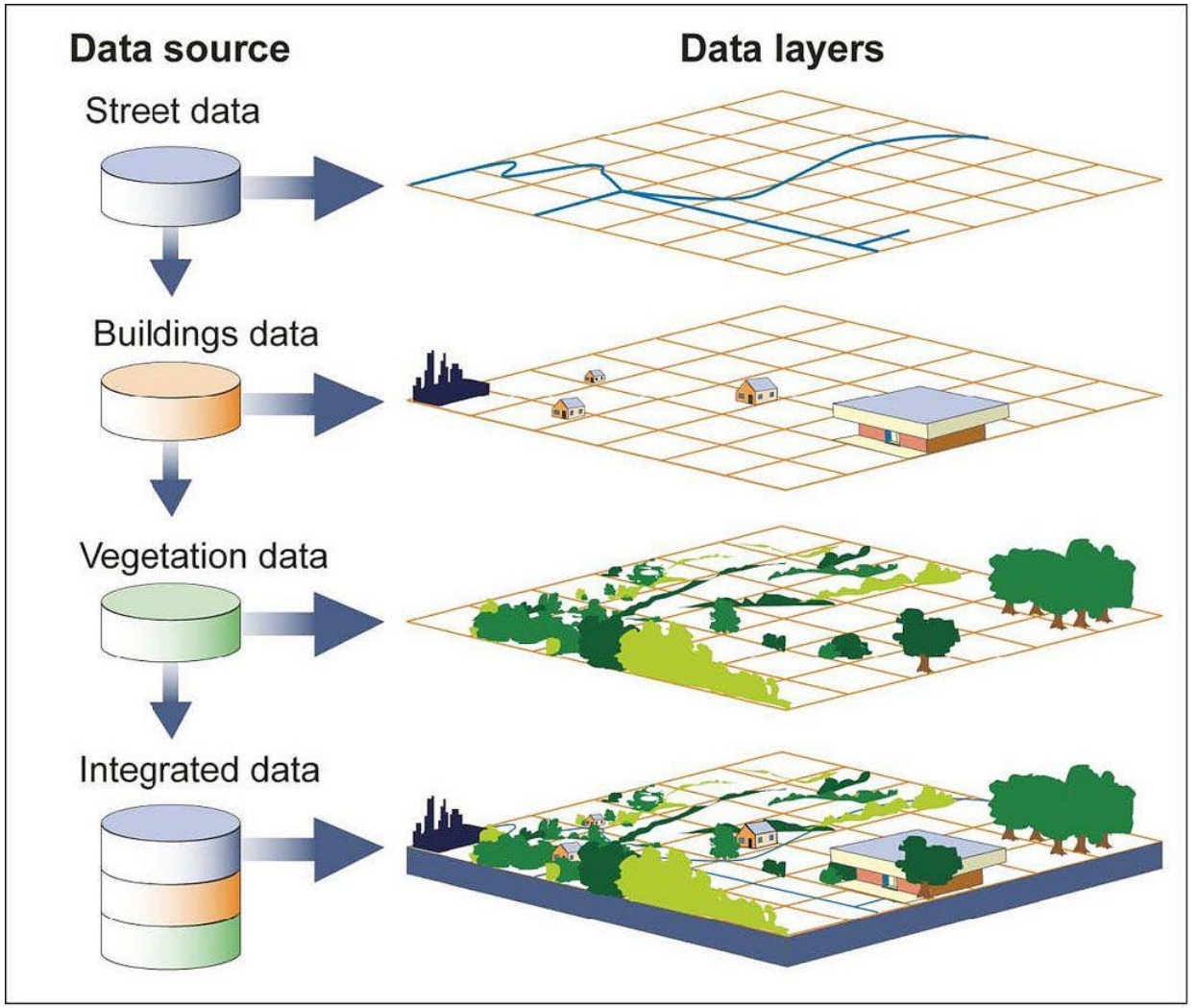


Spatial analysis



Worobey *et al.* 2022

Spatial analysis



Source: GAO.

Cartography and visualisation

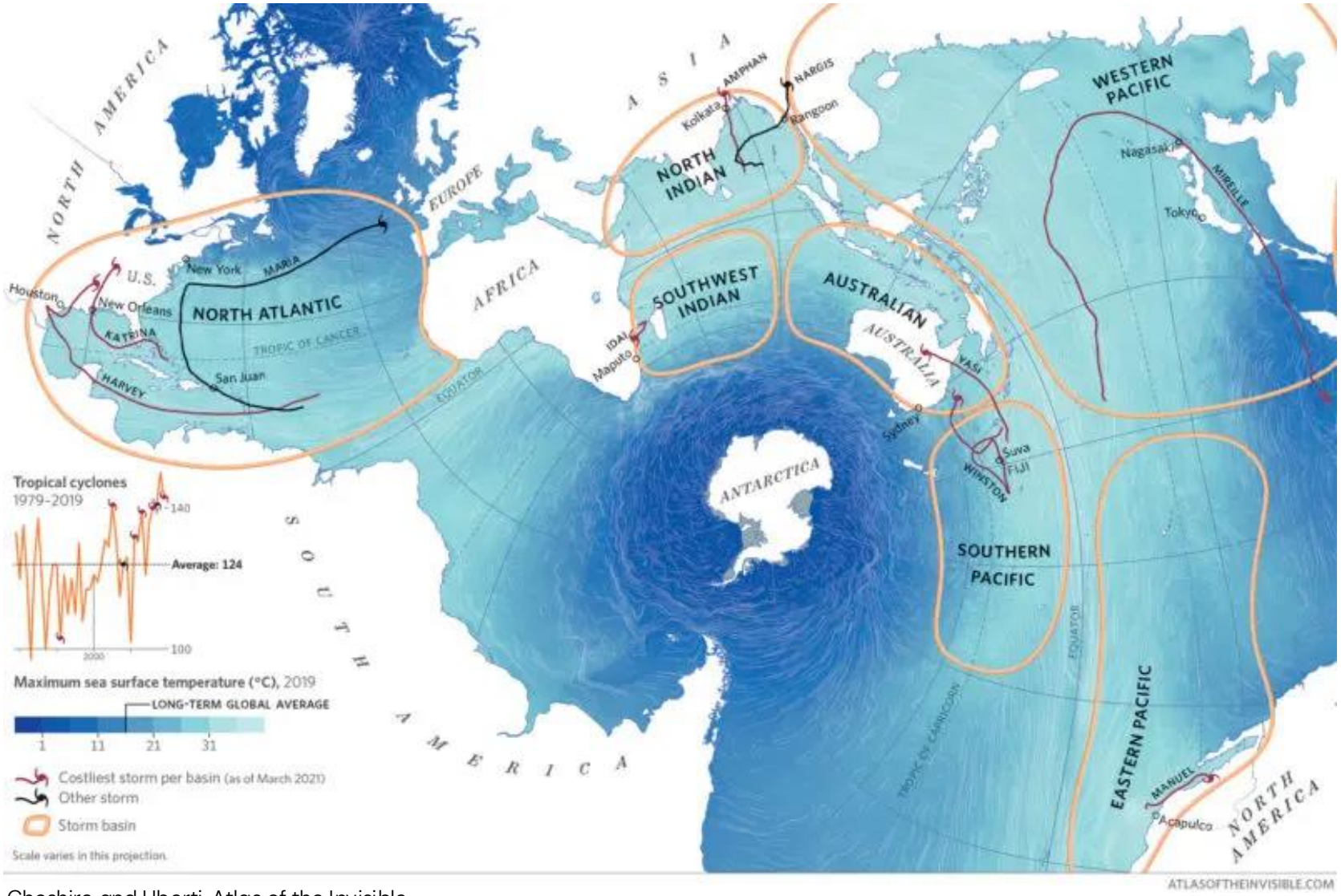


Cartography and visualisation



Cheshire and Uberti. Atlas of the Invisible

Cartography and visualisation



Cheshire and Uberti. Atlas of the Invisible

Choropleth maps

- A choropleth map shades each area according to its value, such as an unemployment rate.
- Numeric values are grouped into classes, and each class gets one colour.
- The classes you choose change what the map shows.

Classification methods

- Equal interval: classes of equal width.
- Quantile: the same number of areas in each class.
- Natural breaks: class boundaries follow gaps in the data.
- Look at the histogram of the variable you are mapping.

Geocomputation

Lots of pillars, but what then is this Geocomputation?

Geocomputation

- Geocomputation means doing GIScience by writing code, so that every step is recorded and can be re-run.
- Scientific research emphasises reproducibility, allowing others to verify results.
- Open-source and programmatic approaches to analysis are increasingly important, partially due to technological advances.
- Advances include access to larger datasets, improved information systems, data mining, AI, and enhanced visualisation techniques.
- Programming languages like R enable reproducible spatial analysis.

Geocomputation

Repeatability	The same methodology will produce the same (or nearly the same) outputs given the same inputs.
Reproducibility	Work can be easily redone or completed by someone else.
Collaboration	Easy to share work and collaborate with others, preferably in real-time, alongside easy integration with version control.
Scalability	Basic: Re-run work easily, adjusting variables and parameters. Intermediate: Expand on work to include larger datasets. Advanced: Suitable for distributed computing.

Geocomputation

“Everyone does need to learn to code. It is no longer sufficient for a GI Scientists to just work with a standard GIS interface: menus, buttons and black boxes.”

Brunsdon and Comber 2021

Conclusion

- A science of Geographic Information is needed to help us create geographic data that can be analysed to turn it into information that can be used as evidence to provide knowledge and thus insight to our biggest – and smallest – geographic challenges and problems.
- Geocomputation emphasises working with geographic data through coding, reproducibility, and modular approaches to deliver robust and transparent analyses.

Questions

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